VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi – 590018



Dashboard for research data analytics

Submitted in partial fulfillment of the requirements as a part of the curriculum,

Bachelors of Engineering in Artificial Intelligence and Machine Learning

Submitted by

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CERTIFICATE

Certified that the Mini Project entitled **"Dashboard for research data analytics"** is presented by

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bonafide student(s) of **CMR Institute of Technology** in partial fulfillment of the requirements as a part of the curriculum, **Bachelors of Engineering in Artificial Intelligence and Machine Learning,** of **Visvesvaraya Technological University, Belagavi** during the year **2024-28**. It is certified that all correction/suggestion indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Entrepreneurial Project report has been approved as it satisfies the academic requirements in respect of the Technical Seminar prescribed for the bachelor of engineering degree.

Dr. Shyam P Joy **(Dr. Shyam P Joy)**

Signature of the Guide **Signature of the HOD**

DECLARATION

We, the undersigned, students of **CMR Institute of Technology**, hereby declare that this mini project report titled

"Dashboard for research data analytics"

is a record of our original work done under the supervision of **Dr. Shyam P Joy,** Head of Department of **AIML**, Department of **CSE (AIML)**. And has not formed the basis of award of any other degree, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

The increasing emphasis on research quality, innovation, patents, entrepreneurship, and conference publications in higher education institutions has created a strong need for centralized academic analytics systems. Educational institutions affiliated with bodies such as All India Council for Technical Education (AICTE) require efficient monitoring and evaluation of faculty research activities to improve institutional rankings, accreditation scores, and academic performance. However, most colleges still rely on fragmented spreadsheets and manual record maintenance, making data analysis and strategic planning difficult.

This project presents a web-based "Research Analytics Dashboard" designed for college faculty and administrators to monitor and analyze departmental performance in the domains of Research, Conference Publications, Innovation, Entrepreneurship, and Patents (RCIEP). The system provides real-time visualization, department-wise comparisons, faculty contribution tracking, and AICTE score-based analytics. It helps institutions identify weak-performing areas and generate actionable insights for improving academic and research outcomes.

The dashboard integrates data management, analytics, and visualization modules into a single platform. Features such as graphical reports, performance trends, score prediction, and recommendation systems assist decision-makers in improving institutional research standards. The proposed system improves transparency, reduces manual effort, and supports data-driven academic planning.

ACKNOWLEDGEMENTS

It is my proud privilege and duty to acknowledge the kind of help and guidance received from several people in preparation of this report. Apart from my own, the success of this report depends largely on the encouragement and guidelines of many others. It would have not been possible to prepare this report in this form without their valuable help, co-operation and guidance.

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I am highly indebted to my project guide **Dr. Shyam P Joy and designation>** for guiding and giving timely advice and suggestions in the successful completion of this project.

My sincere thanks to **<name and designation>,** mini-project coordinator for having supported the work related to this project.

Last but not the least, I would like to put forward my heartfelt acknowledgement to all my classmates, friends and all those who have directly or indirectly provided their overwhelming support during my seminar work and the development of this report.

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1. INTRODUCTION

1.1 Overview

In today's digital academic environment, higher education institutions are increasingly evaluated based on their contributions to research, innovation, patents, entrepreneurship, and conference publications. Regulatory organizations such as All India Council for Technical Education (AICTE) and accreditation bodies focus heavily on research productivity and innovation-driven metrics while assessing institutional quality. As a result, colleges must continuously monitor faculty performance and departmental research activities to remain competitive and improve rankings.

Traditional methods of maintaining faculty achievement records using spreadsheets and manual reports are inefficient, error-prone, and difficult to analyze. Departments often struggle to identify performance gaps, compare research contributions, or evaluate progress against AICTE marking schemes. Without proper analytical tools, administrators cannot make informed strategic decisions for improving institutional performance.

The proposed project, "Dashboard for Research Data Analytics," addresses this challenge by providing a centralized web-based platform for managing and analyzing RCIEP (Research, Conference, Innovation, Entrepreneurship, and Patent) data. The system offers visual dashboards, departmental analytics, faculty performance tracking, and AICTE score evaluation mechanisms. By transforming raw academic data into meaningful insights, the platform helps institutions improve research quality and academic excellence.

1.2 Problem Statement

Most educational institutions maintain research and innovation-related data manually using spreadsheets or disconnected systems. This leads to data inconsistency, lack of real-time analytics, poor visualization, and difficulty in evaluating faculty performance according to AICTE standards. Department heads and administrators face challenges in identifying weak-performing areas and generating institutional reports efficiently.

If this problem is not solved, colleges may experience reduced accreditation scores, inefficient research management, poor strategic planning, and difficulty in improving institutional rankings. Therefore, there is a strong need for an intelligent analytics dashboard that can centralize research data and provide actionable insights.

1.3 Objectives

1. To develop a centralized dashboard for managing RCIEP-related faculty data.
2. To analyze departmental research performance using AICTE marking schemes.
3. To design interactive visualizations and reports for better decision-making.
4. To implement faculty-wise and department-wise performance tracking.
5. To generate recommendations for improving institutional research outcomes.
6. To reduce manual effort in report generation and data management.

1.4 Scope of the Project

The project focuses on managing and analyzing academic research-related activities within a college or university environment. It covers faculty publications, patents, innovation activities, entrepreneurship initiatives, conference participation, and departmental analytics. The system provides visualization dashboards, performance comparison tools, and AICTE score analysis.
The project does not cover plagiarism detection, automated research paper generation, or external indexing database integration such as Scopus or Google Scholar APIs in the current version.

2. LITERATURE SURVEY

1. Research Information Management Systems

Existing Research Information Management Systems (RIMS) are used by universities to store faculty publications and research data. These systems help institutions maintain centralized academic records. However, most systems focus only on storage and reporting rather than advanced analytics and AICTE-oriented performance evaluation.

2. Business Intelligence Dashboards in Education

Educational dashboards developed using tools like Power BI and Tableau provide graphical insights into institutional data. While these tools offer strong visualization capabilities, they often require technical expertise and lack customization for academic research analytics.

3. AI-Based Academic Analytics Platforms

Recent studies show the use of machine learning and predictive analytics in higher education to improve research productivity and student performance. These systems identify trends and forecast institutional growth. However, most solutions focus on student analytics rather than faculty research evaluation.

4. Faculty Performance Evaluation Systems

Several universities use faculty appraisal systems to evaluate publications, patents, and conference contributions. These systems primarily generate static reports and lack intelligent recommendations or comparative analytics aligned with AICTE parameters.

The proposed project addresses these limitations by combining centralized data management, interactive analytics, AICTE score evaluation, and recommendation-based insights in a single platform.

3. METHODOLOGY

3.1 SYSTEM DESIGN

3.1.1 System Requirements

Functional Requirements

1. The system shall allow faculty members to upload RCIEP-related data.
2. The system shall provide department-wise analytics and comparisons.
3. The system shall calculate AICTE-based performance scores.
4. The system shall generate visual reports and charts.
5. The system shall provide role-based authentication for faculty and administrators.
6. The system shall support filtering and searching of records.

Non-Functional Requirements

1. The system should provide secure authentication and data protection.
2. The system should support scalability for multiple departments.
3. The dashboard should be user-friendly and responsive.
4. The system should provide fast report generation and analytics.
5. The system should ensure high availability and reliability.

Hardware and Software Requirements

Category	Development
OS	Windows 10
Processor	Intel Core i5 or above
GPU	NA
RAM	4 GB RAM
Frontend	React.js, Tailwind CSS
Backend	Node.js, Express.js

3.1.2 System Architecture / High-Level Design (HLD)

System Architecture Layers

1. **Client Layer**
    - Faculty Dashboard
    - Admin Dashboard
    - Authentication Interface
2. **Application Layer**
    - Research Analytics Engine
    - AICTE Score Calculator
    - Recommendation Engine
    - Report Generation Module
3. **Database Layer**
    - Faculty Database
    - Publication Records
    - Patent Records
    - Conference Data
    - Innovation and Entrepreneurship Data

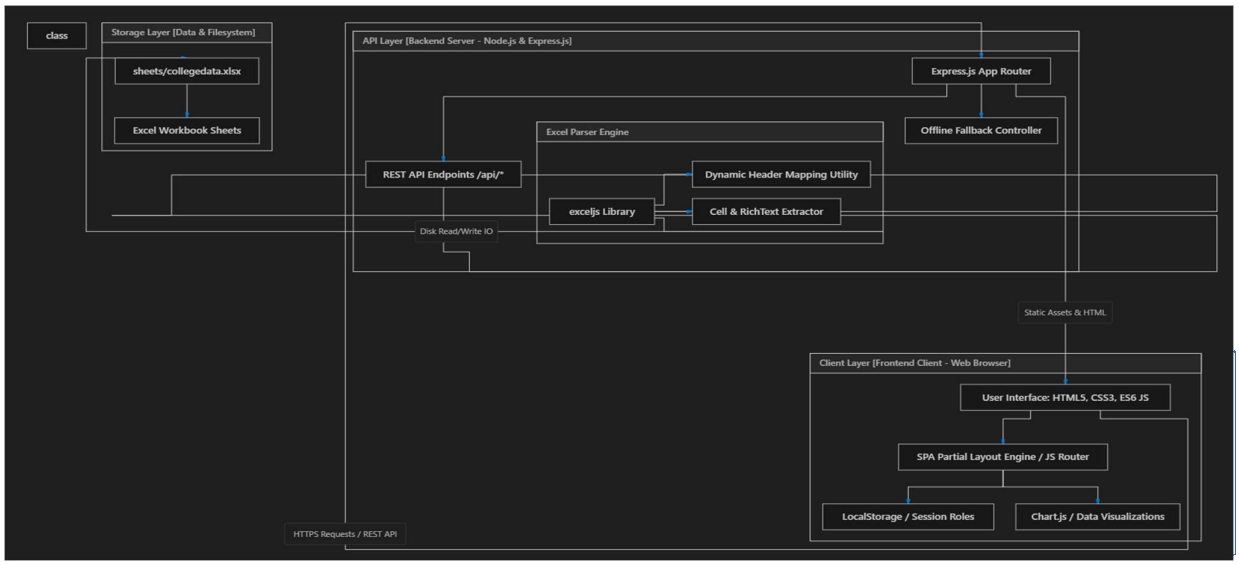


Figure 3.1: System Architecture Diagram

3.1.3 Database Design (ER Diagram / Schema)

Main Tables

1. Faculty
2. Department
3. Publications
4. Patents
5. Conferences
6. Innovation Activities
7. Entrepreneurship Activities

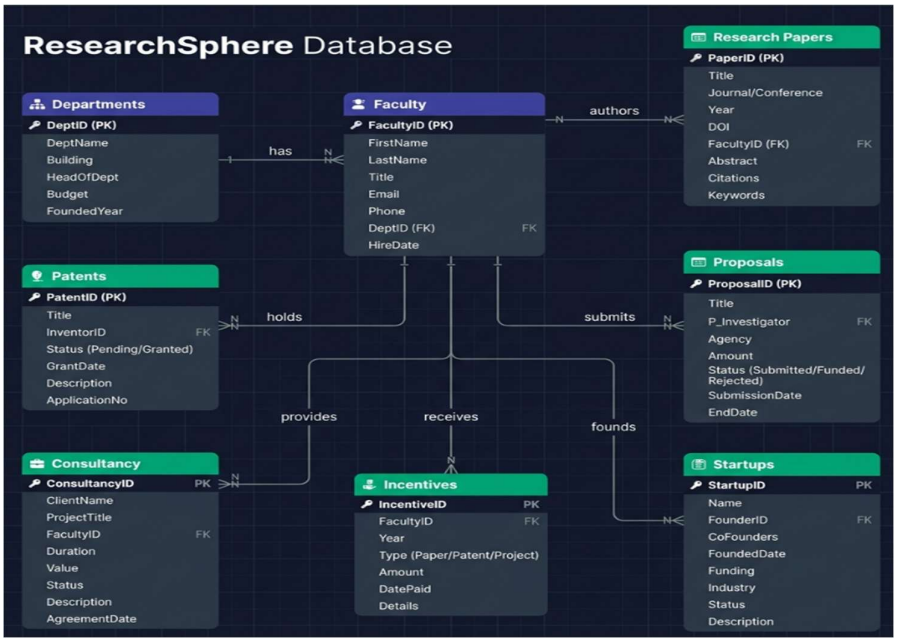


Figure 3.3: Entity Relationship Diagram

3.1.4 Module Description

No.	Module Name	Description
1	User Authentication	Handles secure login and role-based access for faculty and administrators.
2	Data Management Module	Allows users to add, update, and manage RCIEP records.
3	Analytics Engine	Processes data and generates AICTE-based performance analytics.
4	Visualization Module	Displays charts, graphs, and departmental comparisons.
5	Recommendation System	Suggests improvement strategies based on analytics and AICTE criteria.
6	Report Generation Module	Generates downloadable reports for accreditation and administration purposes.

3.2 IMPLEMENTATION

3.2.1 Technology Stack

Layer	Technology / Tool	Reason for Choice
Frontend	React.js	Fast, component-based UI development
Backend	Node.js + Express.js	Efficient REST API development
Database	MySQL	Structured and scalable data storage
Authentication	JWT	Secure authentication mechanism
Deployment	Vercel	Easy deployment and scalability

3.2.2 Implementation of Key Modules

Module 1: User Authentication

The authentication module was implemented using JWT-based authentication with encrypted passwords using bcrypt. Faculty and admin users are assigned role-based access permissions to ensure secure data access.

Functions Used

- Login API
- Register API
- Token Verification Middleware

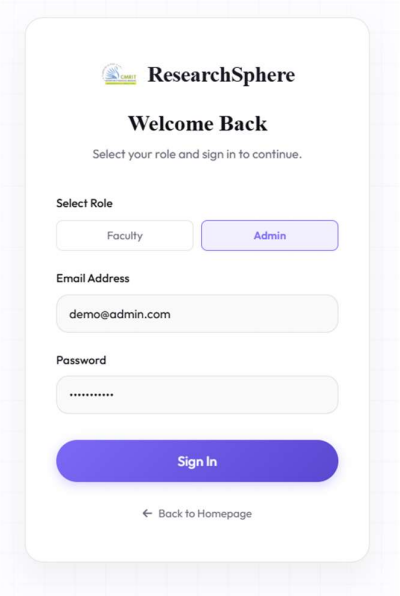


Figure 4.1: [Screenshot of Module 1]

Module 2: Data Management Module

This module allows faculty members to upload publication, patent, conference, and innovation data through forms connected to the backend database. CRUD operations were implemented using REST APIs.

Module 3: Analytics Engine

The analytics engine processes research data and calculates department-wise AICTE scores. Aggregation queries and analytical functions were used to generate insights such as top-performing departments and research trends.

Module 4: Visualization Dashboard

The dashboard displays bar charts, pie charts, line graphs, and heatmaps for institutional analytics. Libraries such as Chart.js and Recharts were used to provide interactive data visualization.

Module 5: Recommendation Engine

The recommendation module identifies weak-performing areas and suggests improvements based on AICTE scoring criteria. For example, it may recommend increasing patent filings or conference participation for a department with low innovation scores.

4. RESULTS & DISCUSSIONS

[Instruction: Testing shows that your system works correctly. Include at least Unit Testing (individual module) and System/Integration Testing (full system). Show test cases in a table and include screenshots of actual results.]

5.1 Testing Methodology

The system was tested using unit testing, integration testing, and user acceptance testing. Individual modules such as authentication, database operations, and analytics generation were tested independently before integrating the complete system. Faculty users and administrators evaluated the usability and accuracy of the dashboard.

5.2 Screenshots of Results

[Instruction: Insert 3–5 screenshots of your working application. Each screenshot should have a caption below it. Cover important screens: home/login page, core functionality output, any graphs/charts if applicable.]

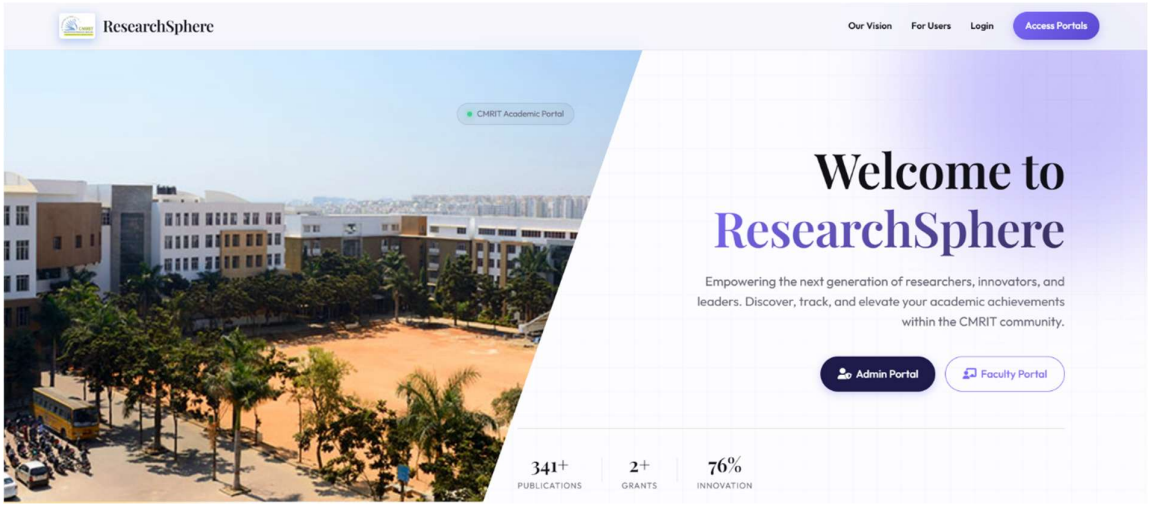


Figure 5.1: **ResearchSphere Home Page**

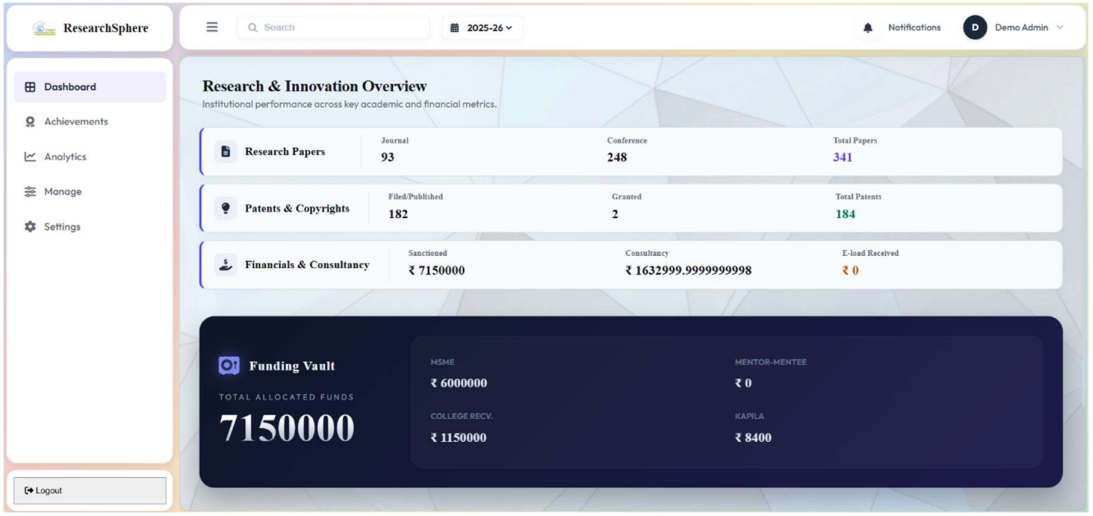


Figure 5.2: landing page of ResearchSphere

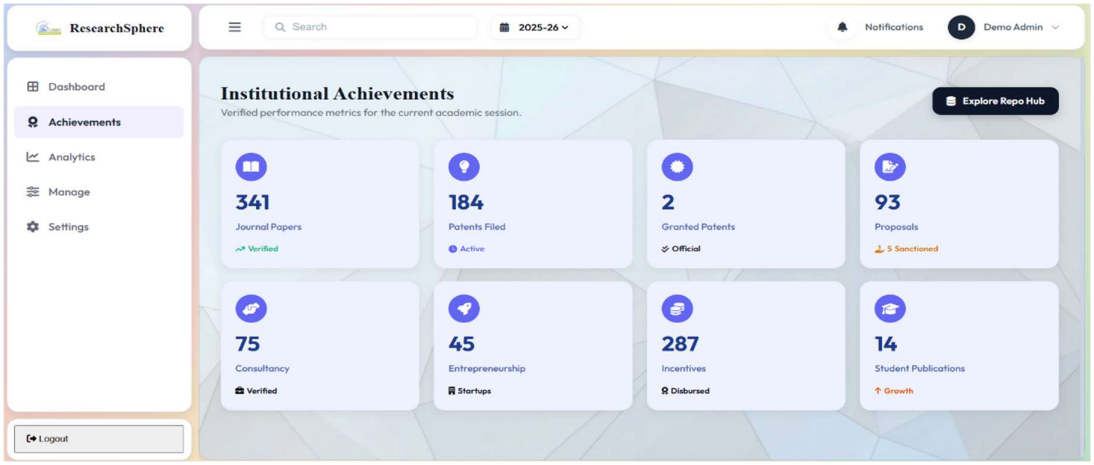


Figure 5.3: achievement section

5.3 Performance Evaluation

[Instruction: If your project involves an ML model, algorithm, or measurable performance, include metrics here. E.g., Accuracy, Precision, Recall, F1-Score for ML projects; or Response Time, Throughput for web apps. A bar/pie chart is a plus. Insert confusion matrix, ROC curve, training curves, or other relevant plots Write 'Not Applicable' if this section does not apply.]

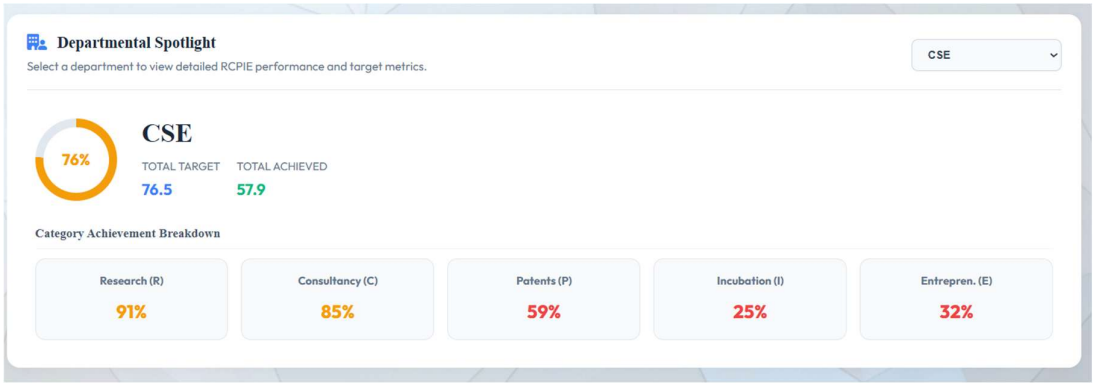


Figure 5.4.: performance evaluation

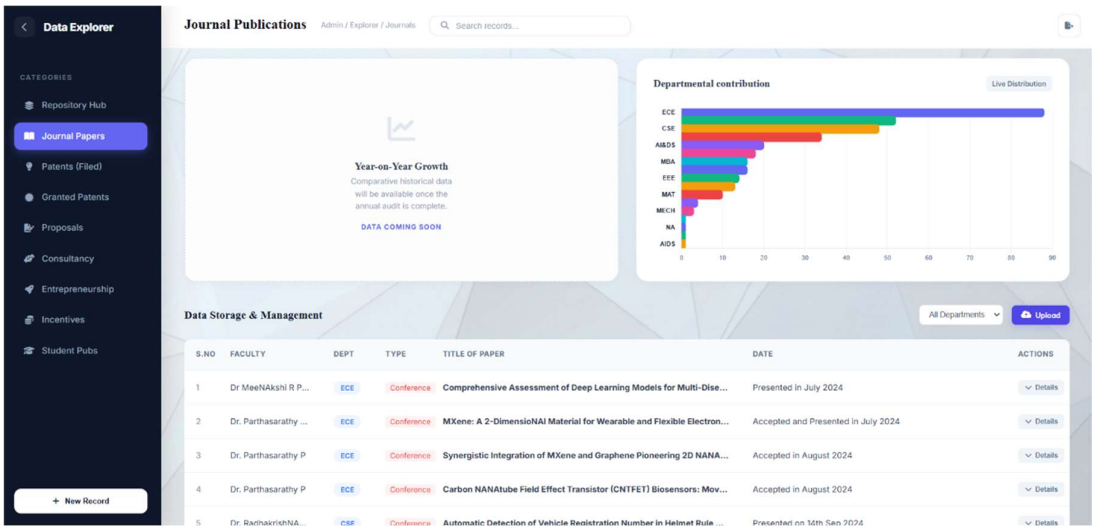


Figure 5.5: data evaluation

5. CONCLUSION

5.1 Conclusion

The project successfully achieved its objective of centralizing RCIEP data analytics into a unified dashboard system. Faculty and administrators could easily track departmental performance and identify improvement areas according to AICTE standards. The analytics and recommendation modules enhanced institutional decision-making and reduced manual workload.

The dashboard also improved transparency and accessibility of research-related data. Visual analytics enabled administrators to compare departments effectively and allocate resources strategically.

5.2 Limitations

The "Research Data Analytics Dashboard" was developed to solve the problem of fragmented and manually managed academic research records. The system successfully integrated data management, analytics, visualization, and AICTE score evaluation into a single web-based platform.

The project met its objectives by providing real-time insights into research, patents, innovation, entrepreneurship, and conference activities. It reduced manual effort, improved transparency, and supported data-driven institutional planning. The project also provided valuable experience in full-stack web development, database design, and data analytics implementation.

5.3 Future Scope

1. Integration with Scopus and Google Scholar APIs.
2. AI-based prediction of departmental research growth.
3. Automated report generation for accreditation agencies.
4. Mobile application development for faculty access.
5. NLP-based analysis of research publication quality.

6. REFERENCES

[Instructions: The references should be numbered serially in the order of their occurrence in the text and their numbers should be indicated within square brackets for e.g. [3]. The section on references should list them in serial order in the following format.

For textbooks

A.V. Oppenheim and R.W. Schafer, Digital Signal Processing, Englewood, N.J., Prentice Hall, 3 Edition, 1975.

For papers

Devid, Insulation design to combat pollution problem, Proc of IEEE, PAS, Vol 71, Aug 1981, pp 1901-1907.]

APPENDIX A: SOURCE CODE

[Instruction: Include the complete, well-commented source code of your project here. Organize it by file or module. Do not paste the code here instead give link to each module in the github.]

GitHub Repository: [https://github.com/<username>/<project-name>]

[Instruction: Give only small snippet of the major file content]

A.1 [Module / File Name: e.g., app.py / main.js / model_train.py]

[add link to file in the github]

A.2 [Module / File Name: e.g., model_utils.py]

[add link to the file in the github]

APPENDIX B: ADDITIONAL SCREENSHOTS

[Instruction: Include additional screenshots here that did not fit in the above chapters. This may include: all UI screens, admin panel views, database records, terminal outputs, model training graphs, etc.]

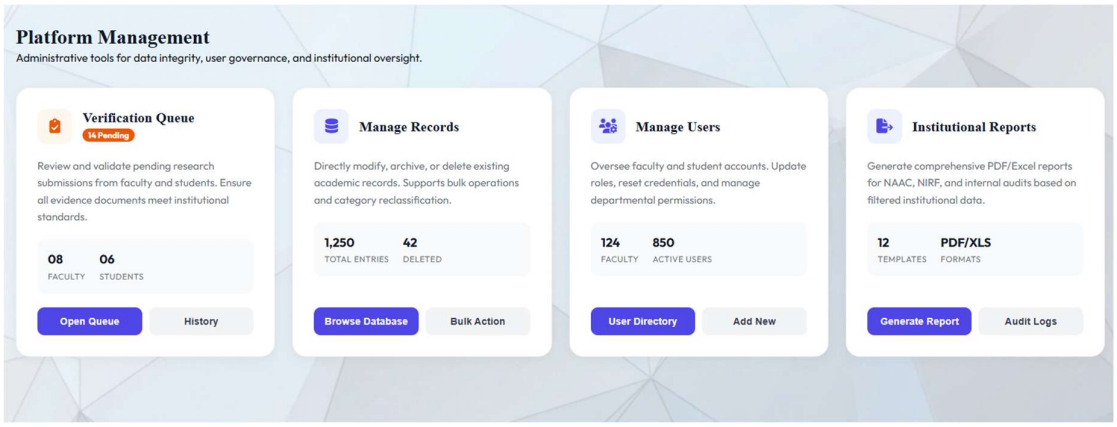


Figure B.1: Features